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Digitalization and supplier environmental performance: An institutional perspective☆

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ABSTRACT

In response to increasing calls for new insights into green supply chain management, we examine the impact of focal firm digitalization on supplier environmental performance. Drawing on institutional theory, we further investigate the moderating effects of three dimensions of institutional pressures faced by focal firms. Utilizing a large-scale dyadic panel dataset comprising 26,954 focal firm-supplier-year observations in China between 2011 and 2022, we employ a two-way cluster-robust standard error estimation to conduct the analysis. Our findings reveal that focal firm digitalization has a positive effect on supplier environmental performance. This effect is more pronounced when focal firms are subject to stronger institutional pressures from external institutional contexts, specifically coercive, mimetic, and normative pressures, proxied by legal system development, industry competition, and overseas sales, respectively. Overall, this study provides novel theoretical and practical insights into how digitalization enhances supplier environmental performance and highlights the key factors influencing its effectiveness.

1. Introduction

Environmental sustainability has evolved from a peripheral corporate concern into a strategic imperative shaping firms' long-term viability, legitimacy, and competitive positioning (Mateska et al., 2023; Song et al., 2024). Nowhere is this shift more consequential than within global supply chains, where environmental impacts are disproportionately concentrated upstream (Porteous et al., 2015). According to the Carbon Disclosure Project (CDP), approximately 90% of corporate carbon emissions originate from supply chain activities, compared to only 10% from focal firms' direct operations (Liu et al., 2024). Consequently, firms are increasingly held accountable not only for their own environmental footprints but also for those of their suppliers (Lo et al., 2018; Shafiq et al., 2017). Recent high-profile cases underscore this interdependence. For example, Apple, often positioned as a sustainability leader, faced significant reputational challenges due to environmental violations by its major supplier, Foxconn Technology Group. Allegations of hazardous waste mismanagement and unsafe handling of toxic materials triggered global scrutiny, compelling Apple to intensify supplier audits, disclose responsibility reports, and enhance supplier training initiatives (Wu, 2024). Such cases illustrate a fundamental governance challenge: how can focal firms effectively influence suppliers' environmental performance across organizational boundaries?.

Traditional supplier management approaches, such as periodic audits, paper-based reporting, and fragmented communication, are frequently reactive, resource-intensive, and limited in transparency (Genovese et al., 2014; Ye et al., 2021). The absence of standardized metrics further complicates cross-supplier comparisons and performance benchmarking. Against this backdrop, digitalization offers new possibilities for transforming supply chain governance. Digital platforms, real-time data analytics, and traceability systems enable continuous monitoring, performance benchmarking, and collaborative problem-solving across supply networks (Li et al., 2020). For instance, Mercedes-Benz's "Ambition 2039" initiative leverages digital technologies to enhance supply chain transparency and environmental accountability. Through digital dashboards, data-driven scorecards, and

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collaborative platforms, the company integrates environmental standards into supplier engagement processes, facilitates traceability of raw materials, and supports capacity-building through digital learning systems. Such initiatives illustrate how digitalization may function not merely as an operational efficiency tool but as a governance mechanism that structures interorganizational sustainability practices.

In the field of operations and supply chain management (OSCM), while scholars have long regarded environmental performance as a core operational metric (Angell & Klassen, 1999; Corbett & Klassen, 2006) and extensively investigated whether the adoption of digital technologies can improve focal firms' environmental performance (Kamble & Gunasekaran, 2023; Park et al., 2023; Ye et al., 2023), we still know little about the extent to which digitalization at the focal firm level generates environmental performance spillovers for upstream suppliers. Addressing this gap is theoretically significant because it reconceptualizes digitalization as a relational capability that shapes sustainability outcomes beyond firm boundaries.

Furthermore, we argue that the magnitude and effectiveness of such influence may be contingent on the institutional environment in which focal firms operate. To provide a more context-sensitive understanding of this relationship, we draw on institutional theory, which emphasizes the role of institutional environments in shaping organizational behaviors (Kauppi, 2013). Institutional pressures are typically categorized into three types: Coercive pressures arise from formal rules and regulations imposed by governments or powerful stakeholders, compelling organizations to conform to legal or regulatory standards. Mimetic pressures are driven by competitive uncertainty and refer to the tendency of organizations to imitate peers or industry leaders that are perceived as successful. Normative pressures originate from societal expectations, professional norms, or market-level demands for legitimacy, especially in international markets with heightened sustainability awareness (Liu et al., 2010; Turkulainen et al., 2017; Villena & Dhanorkar, 2020).

In the context of supply chain digitalization and environmental management, we posit that these institutional pressures systematically shape how focal firms deploy and leverage digital tools in engaging with suppliers. Specifically, coercive pressures, such as stringent environmental regulations, encourage focal firms to adopt digital systems that facilitate compliance tracking and enforcement across the supply chain. Mimetic pressures arising from competitive industry environments may drive firms to proactively use digital platforms to manage supplier sustainability in an effort to match or outperform rivals. Normative pressures, particularly in export-oriented firms operating in markets with strong environmental expectations, push firms to extend digital sustainability standards to upstream partners in order to maintain legitimacy and market access (Zhu et al., 2012). These pressures thus operate at distinct levels, regulatory (coercive), competitive (mimetic), and market-driven (normative), to condition the extent and efficacy of focal firms' digital sustainability initiatives. Based on this theoretical framing, the study addresses two interrelated research questions:

*RQ*1: How does focal firms' digitalization affect their suppliers' environmental performance?

*RQ*2: How do institutional pressures related to legal systems, industry competition, and overseas income that focal firms face influence the relationship between focal firms' digitalization and suppliers' environmental performance?

To examine these questions, we construct a longitudinal dyadic panel dataset comprising 26,954 focal firm-supplier-year observations involving 2,160 focal firms and 2,363 suppliers in China from 2011 to 2022. Using two-way cluster-robust estimations, we find that focal firm digitalization significantly enhances supplier environmental performance. Furthermore, this effect is strengthened when focal firms operate in regions with more developed legal systems, face higher industry competition, and derive a larger share of revenue from overseas markets. A series of robustness checks support the validity of these findings and mitigate endogeneity concerns.

This study contributes to the OSCM literature in three important ways. First, we extend digitalization research beyond intraorganizational outcomes by theorizing digital transformation as an ESG-oriented governance capability that generates cross-boundary environmental spillovers. Second, we advance institutional theory by demonstrating how distinct institutional pressures serve as boundary conditions shaping the effectiveness of digitally enabled interorganizational governance. Third, we integrate digitalization and institutional perspectives to offer a context-contingent explanation of sustainable supply chain governance, highlighting that environmental benefits of digital transformation depend critically on institutional embeddedness. Collectively, our findings provide a theoretically grounded and empirically substantiated account of how digitalization drives environmental upgrading across supply networks, thereby advancing research at the intersection of digital transformation, institutional theory, and sustainable operations management.

2. Related literature and hypotheses development

2.1. Institutional theory and sustainability in supply chains

Institutional theory explains why focal firms increasingly face heightened expectations to monitor and manage environmental practices across their supply base. Firms operate within broader institutional environments composed of regulatory agencies, cultural norms, and societal expectations, which generate coercive, mimetic, and normative pressures that shape organizational behavior as firms seek legitimacy and long-term viability (DiMaggio & Powell, 1983; Mena & Schoenherr, 2023). Within the sustainable supply chain management literature, institutional theory has been widely employed to explain how environmental expectations diffuse across organizational boundaries and motivate stakeholders to adopt practices such as carbon disclosure, supply chain transparency, and sustainable product innovation (Castro-Lopez et al., 2023; Danese et al., 2019; Villena & Dhanorkar, 2020; Wen et al., 2023; Xu et al., 2022). A central implication of this perspective is that focal firms bear disproportionate responsibility for the environmental performance of their supply base. In particular, increasingly stringent international regulations and global sustainability frameworks require firms not only to disclose their own emissions but also to account for environmental impacts arising from upstream suppliers (Wilhelm et al., 2016). Thus, while institutional pressures do not directly prescribe how focal firms govern their suppliers, they establish overarching expectations that incentivize focal firms to adopt more effective governance mechanisms to manage environmental performance across the supply chain. Building on this foundation, our study draws on institutional theory as a contextual rationale for why focal firms must engage in active supplier governance. We then examine how digitalization functions as a governance mechanism that enables focal firms to respond to these institutional expectations through enhanced monitoring, coordination, and capability-building with their suppliers.

2.2. Governance pressures from focal firms and the need for digital enablement

Suppliers' environmental performance is shaped not only by external regulations but also by governance practices imposed by dominant supply chain partners. Focal firms, due to their structural power and central position in supply networks, increasingly exercise governance functions that require suppliers to comply with sustainability standards, adopt green technologies, and avoid environmental misconduct (Mena & Schoenherr, 2023). Environmental incidents at supplier sites, regardless of geographic distance, can generate reputational, regulatory, and financial consequences for the focal firm (Lo et al., 2018). Consequently, focal firms are motivated to build a more transparent and resilient supply chain that mitigates risks across multiple tiers, including environmental risks (Wilhelm et al., 2016), social responsibility risks (Zhang et al., 2024), and financial risks (Ghadge et al., 2021).

However, governing suppliers is inherently challenging. Suppliers often operate in heterogeneous institutional contexts, possess varying capabilities, and may lack the resources necessary to meet environmental expectations (Wilhelm et al., 2016). Traditional governance mechanisms, such as periodic audits or contractual clauses, are insufficient for ensuring real-time visibility and consistent environmental compliance across geographically dispersed suppliers. Digitalization improves the traditional governance mechanism by reducing information asymmetry, supporting real-time monitoring, and fostering interactive coordination. Thus, instead of conceptualizing digitalization as a manifestation of institutional pressure, we view it as a strategic and operational governance mechanism that allows focal firms to more effectively translate institutional expectations into concrete governance actions directed toward suppliers. Under this circumstance, environmental management is not a unilateral effort undertaken solely by focal firms but rather a multi-tier process that involves extensive coordination with supply chain partners and spans various production activities (Sarkis et al., 2010).

2.3. Focal firm digitalization and supplier environmental performance

Digitalization refers to the extent to which a firm integrates disruptive technologies such as artificial intelligence (AI), blockchain, cloud computing, the Internet of Things (IoT), and robotic automation into its strategic and operational processes (Choi et al., 2022; Dalenogare et al., 2018; Meindl et al., 2021). While digital transformation has been shown to improve focal firm sustainability outcomes (Chen et al., 2025; Fang & Li, 2024; Lerman et al., 2022; Meier et al., 2023), less is known about its governance function in facilitating environmental spillovers to suppliers.

From a governance perspective, digitalization equips focal firms with enhanced capabilities to oversee and influence supplier practices. At the strategic level, digital platforms, IoT-enabled monitoring, blockchain-based traceability, and data analytics provide unprecedented visibility and accuracy in assessing supplier environmental performance (Benitez et al., 2022; Nasiri et al., 2020). These technologies strengthen coercive governance by enabling continuous performance tracking, more rigorous auditing, and data-driven compliance enforcement. For instance, Apple's Supplier Clean Energy Program mandates specific renewable energy usage targets and requires suppliers to submit digital reports for auditing and benchmarking, illustrating how digitalization operationalizes coercive control mechanisms in practice (Zimmer et al., 2016). Thus, we argue that focal firm digitalization functions as a coercive governance mechanism that drives suppliers to align with downstream sustainability expectations, even in the absence of formal regulation (Talluri & Sarkis, 2002).

At the operational level, digital technologies facilitate collaborative workflows and knowledge integration across buyer–supplier boundaries (Danese et al., 2019; Frank et al., 2019; Meindl et al., 2021; Mena & Schoenherr, 2023). Base-layer technologies (e.g., cloud computing, blockchain, big data analytics) provide a common infrastructure for data integration, while front-end technologies (e.g., robotics, 3D printing, augmented reality) enhance process control and environmentally efficient practices (Choi et al., 2022; Dalenogare et al., 2018). For instance, cloud-based information systems, digital twin applications, and virtual collaboration tools promote mimetic governance by allowing suppliers to observe and emulate the digital sustainability practices of technologically advanced focal firms (Danese et al., 2019; Laari et al., 2016; Zhang et al., 2025). Meanwhile, frequent digital interactions foster normative governance, creating shared expectations and common environmental standards within cross-organizational routines (Mena & Schoenherr, 2023; Rao, 2002).

In summary, digitalization enhances focal firms' ability to implement coercive, mimetic, and normative governance mechanisms that collectively improve suppliers' environmental outcomes. Rather than acting as institutional pressure per se, digitalization enables focal firms to effectively translate external sustainability expectations into concrete governance practices directed at suppliers.

H1: Focal firm digitalization has a positive effect on supplier environmental performance.

2.4. The moderating effects of institutional pressures

While focal firm digitalization enhances supplier environmental performance by strengthening governance capabilities, the extent to which these digital governance mechanisms are actively deployed and enforced depends on the institutional context in which the focal firm operates. Institutional theory suggests that firms facing stronger external pressures are more likely to intensify internal governance efforts to secure legitimacy and manage sustainability-related risks (DiMaggio & Powell, 1983). Accordingly, institutional pressures do not directly affect suppliers, but shape focal firms' incentives to leverage digital technologies as governance tools for supplier oversight and coordination. Building on prior research, we examine three dimensions of institutional pressure (i.e., coercive, mimetic, and normative pressures) and theorize how each strengthens the effectiveness of focal firm digitalization in improving supplier environmental performance (Yang et al., 2019; Zhu & Sarkis, 2007).

The legal system reflects the strength of formal regulations, enforcement mechanisms, and legal accountability related to environmental protection (Scott, 2008). A well-developed legal system imposes stricter regulatory scrutiny on firms and increases the potential costs of non-compliance, including penalties, reputational damage, and operational disruptions (Campbell, 2007; Hartmann & Uhlenbruck, 2015; Li et al., 2023). Under such conditions, focal firms face heightened responsibility not only for their own environmental performance but also for environmental misconduct occurring within their supply chains.

In response to stronger coercive pressure, focal firms are more likely to reinforce supplier governance to ensure compliance with environmental regulations and reporting requirements. Digital technologies, such as blockchain traceability, IoT monitoring devices, and digital auditing platforms, significantly enhance focal firms' ability to exercise coercive governance by enabling continuous monitoring, standardized reporting, and timely detection of environmental risks at the supplier level. When the legal system is more developed, focal firms have stronger incentives to fully utilize these digital governance tools, thereby amplifying the positive impact of digitalization on supplier environmental performance. Therefore, the governance-enhancing effect of digitalization becomes more pronounced under stronger legal institutional environments.

H2: The development level of the legal system strengthens the positive effect of focal firm digitalization on supplier environmental performance.

Mimetic pressure arises when firms imitate the practices of successful peers to maintain legitimacy and competitive parity, particularly under conditions of uncertainty and intense rivalry (Escobar & Vredenburg, 2011; Villena & Dhanorkar, 2020). In highly competitive industries, firms are more likely to adopt advanced sustainability practices as a means of differentiation and strategic positioning (Castro-Lopez et al., 2023; Galbreth & Ghosh, 2013). As environmental performance increasingly becomes a visible dimension of competitive advantage, focal firms face stronger pressure to demonstrate leadership in sustainable supply chain management.

Digitalization facilitates the diffusion and replication of sustainability-oriented governance practices across firms by increasing transparency, comparability, and benchmarking opportunities. Through digital platforms, focal firms can observe industry leaders' approaches to supplier environmental management and implement similar governance structures within their own supply networks. As industry competition intensifies, focal firms are more motivated to deploy digital technologies to coordinate, monitor, and improve supplier environmental practices, thereby reinforcing mimetic governance effects.

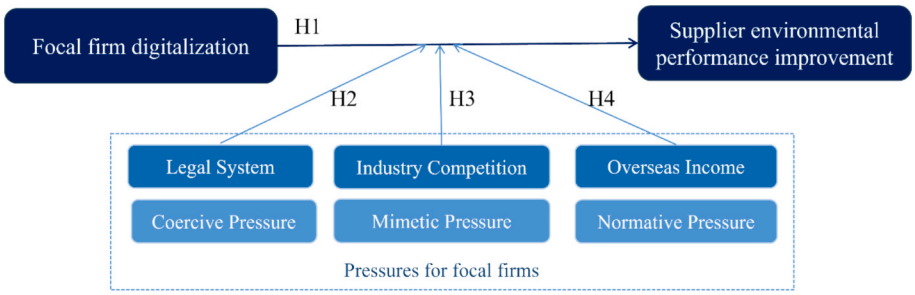

Fig. 1. Conceptual model.

Fig. 2. The impact of focal firm digitalization on supplier environmental performance.

Consequently, the relationship between focal firm digitalization and supplier environmental performance becomes stronger under higher levels of industry competition.

H3: Industry competition strengthens the positive effect of focal firm digitalization on supplier environmental performance.

Normative pressure reflects shared values, professional standards, and societal expectations that define appropriate organizational behavior (DiMaggio & Powell, 1983; Villena & Dhanorkar, 2020). Focal firm demand and preferences are recognized as a critical source of normative pressure, encouraging suppliers to adopt sustainable practices such as circular economy initiatives, environmentally responsible production, and green innovation (Chu et al., 2018; Galbreth & Ghosh, 2013; Govindan & Hasanagic, 2018). Moreover, focal firms with greater exposure to overseas markets face stronger normative expectations regarding environmental responsibility, transparency, and compliance with international standards (Gong et al., 2019). These expectations are often transmitted to suppliers through focal firm requirements, certification systems (e.g., ISO 14001), and sustainability reporting frameworks.

To respond to such normative pressure, focal firms must ensure that their suppliers adhere to globally accepted environmental practices. Digital technologies support this process by enabling standardized data collection, cross-border information sharing, and verifiable sustainability reporting across supply chain partners (Frank et al., 2019; Meindl et al., 2021). When overseas market exposure is higher, focal firms face stronger incentives to rely on digital governance mechanisms to align supplier environmental performance with international norms. As a result, the positive effect of focal firm digitalization on supplier environmental performance is further strengthened.

H4: Overseas market exposure strengthens the positive effect of focal firm digitalization on supplier environmental performance.

Fig. 1 is the conceptual model. Grounded in institutional theory, we present the above hypotheses as depicted in Fig. 2.

3. Methodology

3.1. Sample and data

Our research sample consists of Chinese firms publicly listed on the Shanghai and Shenzhen Stock Exchanges. To construct our focal firm–supplier dyadic data, we utilize information from the FactSet Revere database (hereafter FactSet), which has been widely employed in prior research (Agca et al., 2022; Geng et al., 2022; Song et al., 2024). Measures of focal firm digitalization are obtained from the Chinese Research Data Services Platform (CNRDS), which develops a digitalization index for each listed Chinese firm based on the frequency of digitalization-related keywords disclosed in annual reports. We collect environmental performance data for listed suppliers from the WIND database (Lin et al., 2023; Wang et al., 2024). Additional firm-level financial and control variables are obtained from the China Stock Market and Accounting Research (CSMAR) database, the most commonly used source for Chinese financial data (Li et al., 2023). By integrating these data sources, we construct a comprehensive dataset comprising 26,954 focal firm-supplier dyadic observations, involving 2,363 suppliers and 2,160 focal firms over the period 2011–2022. Notably, a given firm may serve as a focal firm in some dyads and as a supplier in others. Table 1 presents the industry distribution of focal firms and suppliers, respectively. Among all observations, 73.7% of suppliers and 68.3% of focal firms operate in the manufacturing sector. In addition, 85.3% of the dyads involve at least one firm from a manufacturing-related industry.

3.2. Measures

3.2.1. Focal firm digitalization (Digitalization)

Following previous studies (Yang et al., 2023; Zheng, 2024), We construct the independent variable based on the frequency of digitalization-related keywords appearing in annual reports. The CNRDS has systematically collected and structured such digitalization information from listed firms' annual reports since 2007 through advanced textual analysis techniques. The development of this dataset draws on more than 100 relevant keywords, including terms such as intelligent warehousing, intelligent healthcare, intelligent transportation, intelligent banking, industrial internet, deep learning, natural language processing, robotic automation, cloud platform, and digital currency. To address potential heteroskedasticity and reduce skewness in the distribution, we use the natural logarithm of keyword frequency as the operational measure of focal firm digitalization.

Table 1
Distribution of sample firms across industries.

Panel A: Distribution of focal firms across industries

CSRC industry code	Industry	Frequency	Percentage (%)
A	Agriculture, forestry, animal husbandry and fishery	42	0.16
B	Mining	1,783	6.61
C	Manufacturing	18,404	68.28
D	Electricity, heat, gas and water production and supply	1,129	4.19
E	Construction	1,169	4.34
F	Wholesale and Retail	1,308	4.85
G	Transportation, warehousing and postal services	936	3.47
H	Accommodation and catering	19	0.07
I	Information transmission, software and information technology services	1,248	4.63
J	Finance	1	0.00
K	Real estate	414	1.54
L	Leasing and business services	109	0.4
M	Scientific research and technical services	116	0.43
N	Water conservancy, environment and public facilities management	95	0.35
P	Education	14	0.05
Q	Health and social work	30	0.11
R	Culture, sports and entertainment	98	0.36
S	Comprehensive	39	0.14
Total sample size		26,954	100.00

Panel B: Distribution of suppliers across industries

CSRC industry code	Industry	Frequency	Percentage (%)
A	Agriculture, forestry, animal husbandry and fishery	46	0.17
B	Mining	1,384	5.13
C	Manufacturing	19,877	73.74
D	Electricity, heat, gas and water production and supply	578	2.14
E	Construction	635	2.36
F	Wholesale and Retail	747	2.77
G	Transportation, warehousing and postal services	667	2.47
H	Accommodation and catering	9	0.03
I	Information transmission, software and information technology services	1,946	7.22
K	Real estate	112	0.42
L	Leasing and business services	260	0.96
M	Scientific research and technical services	254	0.94
N	Water conservancy, environment and public facilities management	213	0.79
P	Education	10	0.04
Q	Health and social work	16	0.06
R	Culture, sports and entertainment	123	0.46
S	Comprehensive	77	0.29
Total sample size		26,954	100.00

3.2.2. *Supplier environment performance (Escore)*

The Huazheng ESG Rating Index, obtained from the WIND database and widely used in ESG-related research (Deng et al., 2023; Wang et al., 2024), comprises three distinct components: environmental, social, and governance ratings. In this study, we use the environmental dimension to proxy the environmental performance of supplier firms. This score reflects a broad spectrum of environmental outcome indicators, including those related to climate change mitigation, resource utilization, environmental pollution control, and corporate initiatives in environmental management, such as the adoption of eco-friendly strategies and the acquisition of environmental certifications. The score ranges from 0 to 100, with higher values indicating stronger environmental performance.

3.2.3. *Moderators*

To examine the moderating effects of institutional pressures on the relationship between focal firm digitalization and supplier environmental performance, we consider three types of pressure exerted on the focal firm: At the macro level, we measure coercive institutional pressure using the legal system development index (*c_legal_sys*) from the China Marketization Indices. This index assesses the relative comprehensiveness and maturity of legal systems across 31 Chinese provinces and has been widely adopted in prior research (Zhou et al., 2024). A more developed legal system implies stricter enforcement of environmental regulations, thereby compelling firms to adopt more sustainable practices. At the industry level, mimetic pressure is captured by the intensity of industry competition (*c_ind_competition*) faced by the focal firm. Following Chen et al. (2021) and Sun et al. (2018), we calculate this measure as one minus the Herfindahl-Hirschman Index (HHI), where the HHI is computed by summing the squared market shares of all firms within the same industry. A higher value of this variable indicates a more competitive industry environment, in which firms may be more likely to imitate peers' digital and environmental strategies to maintain legitimacy and competitiveness. At the firm level, we proxy normative pressure using the proportion of overseas income (*c_OI*) of the focal firm, calculated as the ratio of overseas revenue to total annual sales. This measure reflects the extent to which firms are embedded in international markets, where environmental norms and expectations are typically more stringent (Chen et al., 2021). Firms with higher overseas exposure are therefore more likely to adopt sustainability practices to meet the expectations of foreign stakeholders.

3.2.4. *Control variables*

We include a comprehensive set of control variables at the supplier, focal firm, and dyadic levels to account for potential sources of heterogeneity that may influence supplier environmental performance. Specifically, at the supplier level, we control for firm size (*S_Firmsize*) and Tobin's *q* as firms with larger total assets or better financial performance may be willing to take more corporate social responsibilities and they may also have more resources to deal with environmental management issues which might not be related to profit. Firm size is measured by the natural logarithm of a firm's annual total assets (Liang et al., 2024). We also control for capital intensity (*Capt_Intensity*), which is measured as total annual assets to annual sales. Firms with high capital intensity may have a higher level of production and distribution activities, generating more potential environmental issues (Song et al., 2023). We control for the cash ratio (*S_Cash*) since dealing with environmental issues should consider the financial constraints (Jia et al., 2025), measured by the ratio of cash assets to total assets. We control for firm growth from the sales growth perspective (*S_Salesgth*), calculated by the sales growth divided by annual sales from the previous year. Firms with a high growth rate may generate more emissions due to a high intensity of production activities. We also control for research and development expenditure for each supplier (*S_RD*), captured as annual research and development expenditure over the annual sales. A higher intensity of R&D or a higher level of innovation capability implies that firms can extend their technology and knowledge to address environmental issues more effectively (Song et al., 2023). We control whether a supplier owns the ISO9001 certificate (*S_ISO9001*), which presents a comprehensive and standard quality management system, and whether it is a state-owned enterprise (*S_SOE*) with dummy variables. Finally, we include the lagged environmental performance score (L.Escore) to account for firm-specific characteristics that persist over time and may influence current environmental outcomes (Chang et al., 2022; Chen et al., 2021).

At the focal firm level, we include analogous control variables to account for similarities and interdependencies between supply chain partners. Thus, we control for the firm size (*C_Firmsize*), cash ratio (*C_Cash*), sales growth (*C_Salesgth*), and development expenditure

Table 2
Descriptive statistics.

Variable	Mean	SD	Min	Max
Escore	61.93	7.700	31.08	92.30
Digitalization	1.880	1.340	0.000	6.140
S_Firmsize	22.87	1.630	16.41	28.64
Tobin's *q*	1.940	1.720	0.640	106.1
Capt_Intensity	2.170	1.930	0.090	80.47
S_Cash	0.130	0.100	0.000	0.830
C_Cash	0.140	0.100	0.000	0.970
S_Salesgth	0.210	1.690	−1.000	103.8
C_Salesgth	0.400	13.55	−1.000	944.1
C_Firmsize	24.63	1.990	16.41	28.64
S_RD	0.050	0.060	0.000	4.250
C_RD	0.180	16.03	0.000	1860
S_ISO9001	0.310	0.460	0.000	1.000
C_ISO9001	0.240	0.430	0.000	1.000
S_SOE	0.340	0.470	0.000	1.000
C_SOE	0.580	0.490	0.000	1.000
Techproximity	0.280	0.370	0.000	1.120
Same_Ind	0.230	0.420	0.000	1.000
Same_Area	0.190	0.400	0.000	1.000
L.Escore	61.39	8.030	31.08	92.30

(*C_RD*) for each focal firm. We also control for whether a focal firm has a standard quality management system (*C_ISO9001*) and whether it is a state-owned enterprise (*C_SOE*) with dummy variables.

At the dyadic level, we control for technological proximity (*Techproximity*) between each focal firm-supplier pair. A closer technological alignment may increase the likelihood of production and operational activities being transferred from the focal firm to the supplier, potentially leading to higher environmental emissions and associated challenges (Song et al., 2023). We adopt the measurement approach proposed by Song et al. (2023) to quantify this proximity. Additionally, we include two binary variables to capture geographical and industry relatedness: whether the supplier and the focal firm are headquartered in the same province (*Same_Area*), and whether they operate within the same industry sector (*Same_Ind*). Finally, to account for unobserved temporal variation, we incorporate year fixed effects (*Year*) in our models (Agca et al., 2022; Song et al., 2023).

3.3. Identification strategy

Our dyadic panel dataset comprises 26,954 focal firm-supplier-year observations, involving 2,363 suppliers and 2,160 focal firms between 2011 and 2022. Accordingly, multiple dyads may share the same suppliers or focal firms, which can induce cross-dyad correlation and lead to underestimated standard errors. Following prior studies (Cameron et al., 2011; Song et al., 2023), we therefore employ a two-way cluster-robust standard error estimation to address this potential concern. The two-way cluster-robust standard error estimation controls for such multi-level correlation by clustering at the two dimensions separately and excludes the possibility of double counting in calculating variance estimators. Hence, we employ the following model to conduct the regression analysis for testing H1.

$$Escore_{sct} = \beta_0 + \beta_1 Digitalization_{ct} + \beta_2 S_{st} + \beta_3 C_{ct} + \beta_4 D_{sct} + \beta_5 T_t + \epsilon_{sct} \quad (1)$$

where *s*, *c*, and *t* represent supplier firm, focal firm, and year, respectively. The dependent variable is the environment score for each supplier each year ($Escore_{sct}$). $Digitalization_{ct}$ denotes the intensity of focal firm digitalization each year. S_{st}, C_{ct}, D_{sct} represent vectors of control variables accounting for supplier-level, focal firm-level, and dyadic-level heterogeneity, respectively. In addition, we include year fixed effects T_t to control for time-specific shocks.

To test H2 to H4, we include three interaction terms into Eq. (1), respectively.

$$Escore_{sct} = \beta_0 + \beta_1 Digitalization_{ct} \times c_leg_sys_{ct} + \beta_2 S_{st} + \beta_3 C_{ct} + \beta_4 D_{sct} + \beta_5 T_t + \epsilon_{sct} \quad (2)$$

Table 3
Correlation matrix.

	Escore	Digitalization	S_Firmsize	Tobin's *q*	Capt_Intensity	S_Cash	C_Cash	S_Salesgth	C_Salesgth	C_Firmsize
Escore	1									
Digitalization	0.016***	1								
S_Firmsize	0.210***	−0.178***	1							
Tobin's *q*	−0.102***	0.074***	−0.309***	1						
Capt_Intensity	−0.061***	−0.032***	−0.102***	0.069***	1					
S_Cash	−0.003	0.087***	−0.141***	0.177***	0.039***	1				
C_Cash	−0.009	0.138***	−0.045***	0.041***	−0.014**	0.039***	1			
S_Salesgth	−0.027***	0.002	0.003	0.015**	−0.035***	0.012**	−0.001	1		
C_Salesgth	−0.006	−0.011*	0.009	0.022***	−0.001	0.012*	0.013**	0.021***	1	
c_Firmsize	−0.013**	0.067***	−0.134***	−0.003	0.049***	0.008	−0.182***	−0.010*	−0.024***	1
S_RD	0.004	0.176***	−0.226***	0.173***	0.242***	0.187***	0.044***	−0.025***	−0.001	0.054***
C_RD	−0.011*	−0.011*	0.001	0.006	0.007	0.013**	0.040***	−0.001	0.000	−0.016***
S_ISO9001	0.133***	0.032***	−0.085***	0.016***	−0.044***	0.031***	0.003	−0.003	−0.001	0.021***
C_ISO9001	0.003	0.067***	−0.031***	0.006	−0.028***	−0.035***	0.090***	−0.006	−0.006	−0.085***
S_SOE	0.030***	−0.207***	0.472***	−0.156***	−0.031***	−0.018***	−0.086***	−0.021***	−0.002	−0.081***
C_SOE	−0.028***	−0.225***	0.047***	−0.062***	0.069***	−0.009	−0.022***	−0.011*	−0.023***	0.325***
Techproximity	−0.010	0.020***	−0.089***	−0.010	−0.031***	−0.035***	0.041***	−0.003	−0.011*	−0.025***
Same_Ind	−0.048***	0.062***	−0.050***	0.019***	0.010	−0.023***	0.086***	−0.005	−0.009	−0.067***
Same_Area	−0.035***	0.028***	−0.034***	−0.008	−0.005	0.025***	−0.009	−0.004	0.005	−0.044***
L.Escore	0.688***	−0.014**	0.163***	−0.102***	−0.040***	−0.019***	−0.015**	−0.019***	−0.007	0.016***

	S_RD	C_RD	S_ISO9001	C_ISO9001	S_SOE	C_SOE	Techproximity	Same_Ind	Same_Area	L.Escore
S_RD	1									
C_RD	0.000	1								
S_ISO9001	0.082***	−0.006	1							
C_ISO9001	0.006	−0.005	0.030***	1						
S_SOE	−0.240***	−0.007	−0.093***	−0.034***	1					
C_SOE	−0.038***	−0.011*	−0.023***	−0.157***	0.150***	1				
Techproximity	0.033***	−0.006	0.064***	0.100***	−0.090***	−0.076***	1			
Same_Ind	−0.014**	−0.005	−0.029***	0.003	−0.040***	−0.038***	0.078***	1		
Same_Area	0.004	−0.004	0.000	0.041***	0.048***	−0.031***	−0.001	0.004	1	
L.Escore	0.000	−0.013**	0.130***	−0.023***	0.020***	0.003	−0.013**	−0.069***	−0.041***	1

Table 4
Regression results.

	Model 1	Model 2	Model 3	Model 4	Model 5
	Dependent Variable: Supplier environmental performance				
Digitalization		0.105***	−0.303	−0.157	0.046
		(2.60)	(−1.51)	(−1.58)	(0.95)
Legal_System_Development			0.002		
			(0.05)		
Digitalization * Legal_System_Development			0.030**		
			(2.04)		
Industry_Competition				−0.749*	
				(−1.80)	
Digitalization * Industry_Competition				0.328***	
				(2.66)	
Overseas_Income					−0.400
					(−1.04)
Digitalization * Overseas_Income					0.345**
					(2.17)
S_Firmsize	0.605***	0.611***	0.615***	0.617***	0.612***
	(5.30)	(5.38)	(5.59)	(5.42)	(5.40)
Tobin' s *q*	0.034	0.033	0.033	0.035	0.033
	(1.00)	(0.99)	(0.98)	(1.06)	(0.98)
Capt_Intensity	−0.083**	−0.079**	−0.080**	−0.085**	−0.079**
	(−2.28)	(−2.20)	(−2.18)	(−2.29)	(−2.19)
S_Cash	1.690**	1.623**	1.723**	1.717**	1.655**
	(2.15)	(2.06)	(2.20)	(2.15)	(2.11)
C_Cash	−0.236	−0.404	−0.454	−0.440	−0.389
	(−0.61)	(−1.06)	(−1.17)	(−1.13)	(−1.02)
S_Salesgth	−0.058**	−0.058***	−0.058***	−0.047**	−0.058**
	(−2.55)	(−2.60)	(−2.63)	(−2.14)	(−2.56)
C_Salesgth	−0.000	−0.000	−0.000	−0.000	−0.000
	(−0.32)	(−0.21)	(−0.17)	(−0.08)	(−0.19)
C_Firmsize	−0.047*	−0.055**	−0.061**	−0.063**	−0.059**
	(−1.91)	(−2.06)	(−2.22)	(−2.30)	(−2.27)
S_RD	1.549	1.338	1.731	1.453	1.329
	(0.96)	(0.83)	(1.07)	(0.91)	(0.83)
C_RD	−0.002***	−0.001***	−0.001***	−0.001***	−0.001***
	(−6.96)	(−6.30)	(−6.22)	(−5.91)	(−6.28)
S_ISO9001	0.765***	0.768***	0.774***	0.750***	0.758***
	(3.77)	(3.79)	(3.89)	(3.63)	(3.75)
C_ISO9001	0.191*	0.186*	0.156	0.170*	0.162
	(1.94)	(1.87)	(1.61)	(1.70)	(1.64)
S_SOE	−0.297*	−0.284	−0.276	−0.298*	−0.291
	(−1.67)	(−1.60)	(−1.55)	(−1.68)	(−1.64)
C_SOE	−0.263***	−0.205**	−0.203**	−0.185**	−0.164*
	(−3.00)	(−2.35)	(−2.30)	(−2.13)	(−1.85)
Techproximity	−0.089	−0.075	−0.100	−0.090	−0.068
	(−0.76)	(−0.64)	(−0.86)	(−0.77)	(−0.59)
Same_Ind	0.096	0.079	0.090	0.084	0.058
	(0.68)	(0.57)	(0.64)	(0.59)	(0.42)
Same_Area	−0.043	−0.056	−0.096	−0.052	−0.066
	(−0.40)	(−0.53)	(−0.87)	(−0.48)	(−0.62)
L.Escore	0.635***	0.635***	0.639***	0.635***	0.636***
	(40.33)	(40.48)	(41.35)	(40.26)	(40.49)
_cons	10.118***	9.985***	9.773***	10.614***	10.062***
	(4.24)	(4.20)	(4.38)	(3.97)	(4.26)
Observation	26,954	26,954	26,202	26,518	26,954
Year fixed	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.500	0.500	0.505	0.501	0.500
F	181.609	174.565	156.167	158.073	158.652

Notes: *p < 0.1, **p < 0.05, ***p < 0.01. Robust t statistics in parentheses.

$$Escore_{sct} = \beta_0 + \beta_1 Digitalization_{ct} \times c_ind_comp_{ct} + \beta_2 S_{st} + \beta_3 C_{ct} + \beta_4 D_{sct} + \beta_5 T_t + \epsilon_{sct} \quad (3)$$

$$Escore_{sct} = \beta_0 + \beta_1 Digitalization_{ct} \times c_OI_{ct} + \beta_2 S_{st} + \beta_3 C_{ct} + \beta_4 D_{sct} + \beta_5 T_t + \epsilon_{sct} \quad (4)$$

4. Empirical results

4.1. Main result

Tables 2 and 3 exhibit the descriptive statistics and correlation matrix of all variables. The average variance inflation factor (VIF) across all variables is 1.13, with the maximum value being 1.50, suggesting that multicollinearity is not a concern in our analysis. In Table 4, Model 1 serves as the baseline model that only includes control variables. Model 2 subsequently incorporates the independent variable. Models 3 to 5 examine the moderating effects of three types of institutional pressures faced by focal firms: legal system development (*Legal_System_Development*), industry competition (*Industry_Competition*), and overseas income (*Overseas_Income*), respectively.

In model 2, the coefficient of *Digitalization* is significantly positive ($\beta = 0.105, p < 0.01$), indicating that focal firm digitalization significantly enhances supplier environmental performance, therefore supporting H1. In model 3, we investigate the moderating effect of legal system development. The coefficient of interaction term is positively

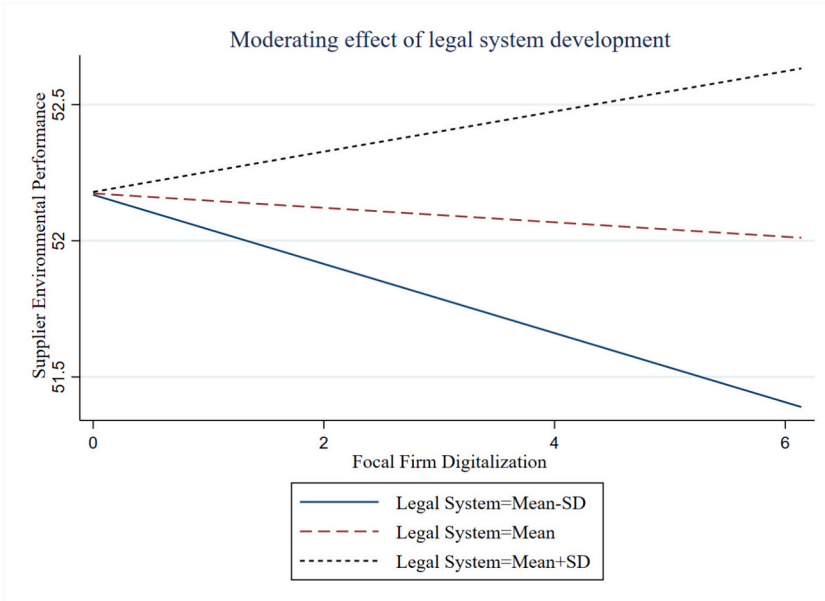

Fig. 3. The moderating effect of legal system development.

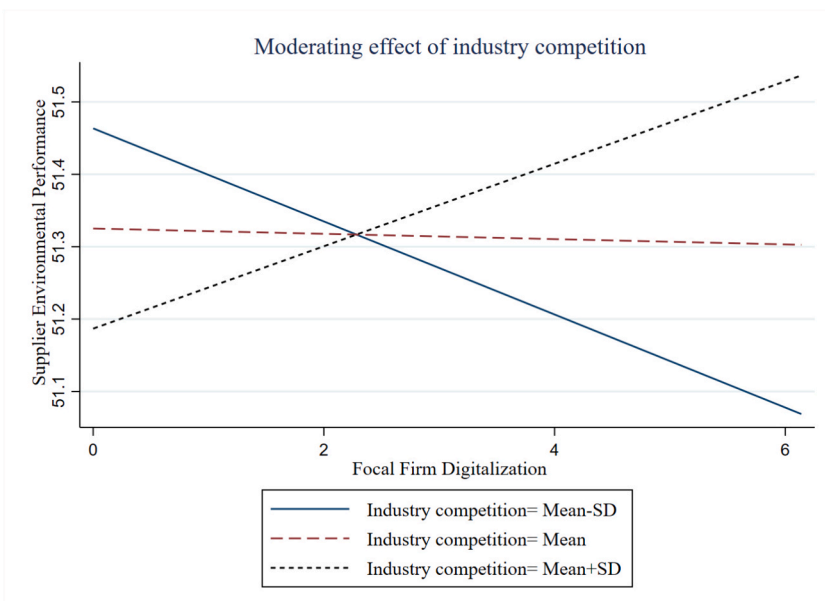

Fig. 4. The moderating effect of industry competition.

significant ($\beta = 0.030, p < 0.05$), indicating that the positive impact of focal firm digitalization is amplified when focal firms are embedded in regions with more developed legal systems. Thus, *H2* is supported. In model 4, the interaction term is positively significant ($\beta = 0.328$, $p < 0.01$). This result indicates that greater industry competition strengthens the positive effect of focal firm digitalization. Specifically, when focal firms operate in more competitive environments, they are more likely to emulate leading practices, thereby leveraging digital technologies more effectively to promote supplier environmental performance. Thus, *H3* is supported. In Model 5, the coefficient of interaction term between *Digitalization* and *Overseas Income* is positively significant ($\beta = 0.345, p < 0.05$), implying that the positive effect of focal firm digitalization is further enhanced when focal firms have greater exposure to international markets and thus face stronger normative pressure from overseas stakeholders. Thus, *H4* is supported. Figs. 3 to 5 graphically illustrates that the positive impact of focal firm digitalization on supplier environmental performance is strengthened by more advanced legal system development, intensified industry competition, and increased overseas income.

4.2. Robustness checks

To test the robustness of our main regression results, we adopt a series of alternative measurements and estimation strategies. As shown in Table 5, Model 1, we begin by winsorizing both the dependent and independent variables to mitigate the influence of extreme values. In Model 2, we introduce additional control variables into the regression. Given that environmental performance is closely associated with corporate governance quality (Walls et al., 2012), we include three governance-related variables to account for the supplier's governance

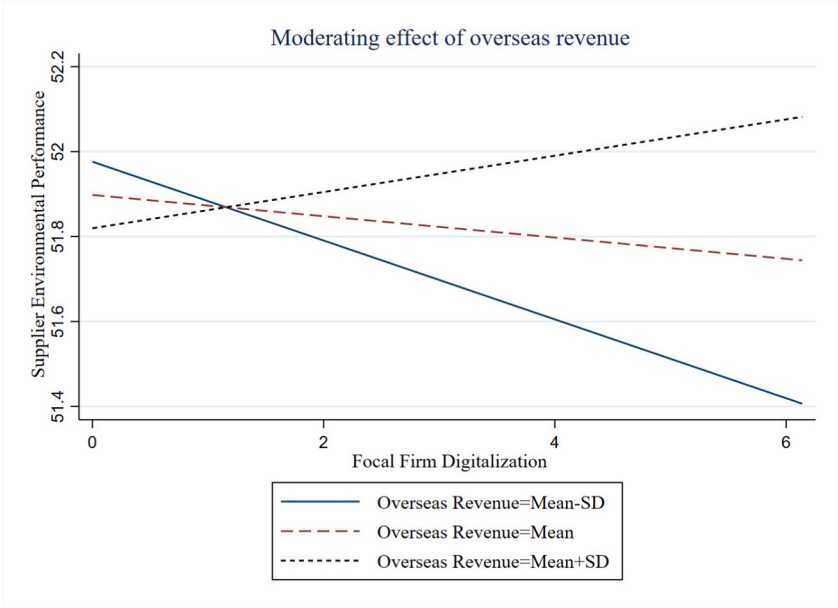

Fig. 5. The moderating effect of overseas revenue.

characteristics: CEO duality (*Duality*), the proportion of independent directors (*Indepdir*), and the proportion of female directors (*Femaledir*). In Model 3, we replace the dependent variable with the supplier's environmental performance in the following year to examine temporal consistency. Model 4 employs an alternative measure of digitalization derived from the CSMAR database, an index calculated based on the digital technology keywords frequency in corporates' annual reports and widely adopted in prior studies (Feng & Zhu, 2024). Finally, Model 5 replaces the two-way cluster-robust standard error estimation with conventional robust standard errors. Overall, the results presented in Table 5 consistently support our main findings.

We further conduct three robustness tests to address potential endogeneity concerns, as reported in Table 6. Although our sample data are collected without ex ante selection bias, the formation of focal firm-supplier relationships may still be influenced by unobservable factors. For example, focal firms that disclose greater levels of digitalization in their annual reports may possess superior capabilities in managing and governing supply chain partners. Furthermore, missing observations in our dyadic dataset may lead to an underestimation of underlying relationships. As such, we employ the propensity score matching method (PSM) to mitigate these selection bias concerns. Specifically, we create a dummy variable *Treatment* that equals zero for observations where *Digitalization* falls within the bottom 25th percentage or equals to one otherwise (Song et al., 2024). We incorporate all control variables as covariate variables in calculating the propensity score and utilize propensity score matching (PSM) to match the treatment group (Treatment = 1) with the control group (Treatment = 0) in a one-to-one, without replacement manner. We employ a caliper distance of 0.01 with the nearest neighboring method (Chang et al., 2022). Subsequently, we re-estimate Model 2 with the matched samples (treatment and control groups) as presented in column 1 of Table 6. The coefficient of *Digitalization* remains positive and significant ($\beta = 0.146, p < 0.01$), which is consistent with our main results.

While PSM addresses concerns arising from observable factors that may bias our estimates, it does not fully eliminate the potential influence of omitted variables and other endogeneity issues. To further mitigate these concerns, we employ an instrumental variable regression using a two-stage least square approach with two-way clustered-robust error estimation (IV-2SLS).

Specifically, we identify the number of IT industry employees in the focal firm's province as an appropriate instrumental variable, as it is strongly correlated with focal firm digitalization while exhibiting no direct association with supplier environmental performance, thereby satisfying the exogenous requirements for an effective IV. The results of the IV-2SLS regression are presented in columns 2 and 3 of Table 6. The coefficient of *it_employee* in column 2 is positively significant ($\beta = 0.154$, $p < 0.01$), indicating a significant correlation between the instrumental variable and our independent variable. Also, the coefficient of *Digit* in column 3 shows significantly positive ($\beta = 0.849$, $p < 0.05$), suggesting the IV regression estimates are significant and highly consistent with the main results. Moreover, we conduct the underidentification test and weak instruments test to examine the validity of this estimation. Firstly, the p value for Kleibergen–Paap rk LM statistic is 0.0024 ($p < 0.01$), indicating a pass in the underidentification test (Kleibergen & Paap, 2006). Secondly, the Cragg-Donald Wald F value is 278.446, larger than the 10% maximal critical values of Stock-Yogo weak ID tests, suggesting that this IV is not weak (Stock & Yogo, 2002).

Finally, to mitigate the potential concern of reverse causality, we re-estimate the regression model, with focal firm digitalization as the dependent variable and supplier environmental performance as an independent variable. The result in column 4 of Table 6 demonstrates that the coefficient of supplier environmental performance is insignificant, indicating that reverse causality is not a major concern in our analysis.

5. Discussion

This study examines the impact of focal firm digitalization on supplier environmental performance and the moderating role of institutional pressures. The findings indicate that focal firm digitalization significantly enhances suppliers' environmental performance. Moreover, this positive effect becomes more pronounced when focal firms face stronger institutional pressures, including coercive, mimetic, and normative pressures.

Table 5
Robustness tests.

	Model 1	Model 2	Model 3	Model 4	Model 5
	Winsorization	New controls	Alternative DV	Alternative IV	Robust error
	Dependent Variable: Supplier environmental performance				
Digitalization	0.095**	0.094**	0.133***	0.005***	0.095***
	(2.332)	(2.286)	(3.649)	(2.836)	(3.378)
S_Firmsize	0.616***	0.678***	0.291***	0.733***	0.616***
	(6.390)	(10.388)	(4.933)	(8.301)	(22.059)
Tobin' s *q*	0.035	0.062	0.029	0.055	0.035**
	(1.082)	(1.582)	(1.065)	(1.593)	(2.266)
Capt_Intensity	−0.074**	−0.080**	−0.027	−0.093**	−0.074***
	(−2.096)	(−2.103)	(−1.398)	(−2.449)	(−4.663)
S_Cash	1.494*	1.241	1.279**	1.267	1.494***
	(1.948)	(1.635)	(2.102)	(1.571)	(4.200)
C_Cash	−0.482	−0.522	0.029	−0.528	−0.482
	(−1.293)	(−1.392)	(0.095)	(−1.216)	(−1.429)
S_Salesgth	−0.053**	−0.053**	−0.012	−0.055**	−0.053***
	(−2.419)	(−2.421)	(−0.896)	(−2.461)	(−3.462)
C_Salesgth	−0.000	−0.001	−0.002**	0.000	−0.000
	(−0.288)	(−0.392)	(−2.572)	(0.003)	(−0.259)
C_Firmsize	−0.057**	−0.054**	−0.064***	−0.098***	−0.057***
	(−2.109)	(−2.002)	(−2.929)	(−3.287)	(−3.056)
S_RD	0.793	0.991	0.963	0.856	0.793
	(0.543)	(0.665)	(0.596)	(0.567)	(1.031)
C_RD	−0.002***	−0.002***	−0.002***	−0.051***	−0.002*
	(−7.090)	(−6.604)	(−5.523)	(−9.419)	(−1.908)
S_ISO9001	0.761***	0.835***	0.235	0.736***	0.761***
	(3.982)	(4.881)	(1.639)	(3.702)	(10.310)
C_ISO9001	0.191*	0.163	0.192**	0.238**	0.191**
	(1.925)	(1.626)	(2.391)	(2.351)	(2.463)
S_SOE	−0.232	−0.263	−0.044	−0.330*	−0.232***
	(−1.327)	(−1.487)	(−0.323)	(−1.885)	(−2.884)
C_SOE	−0.207**	−0.211**	−0.206***	−0.218**	−0.207***
	(−2.364)	(−2.399)	(−2.871)	(−2.304)	(−2.755)
Techproximity	−0.071	−0.078	0.050	−0.023	−0.071
	(−0.610)	(−0.661)	(0.605)	(−0.179)	(−0.820)
Same_Ind	0.088	0.035	0.276***	0.093	0.088
	(0.635)	(0.256)	(2.833)	(0.601)	(1.206)
Same_Area	−0.065	−0.055	−0.042	−0.118	−0.065
	(−0.614)	(−0.509)	(−0.466)	(−0.981)	(−0.787)
L.Escore	0.616***	0.613***	0.779***	0.591***	0.616***
	(48.290)	(50.104)	(61.916)	(41.869)	(122.561)
Duality		−0.127			
		(−0.702)			
Indepdir		−0.572			
		(−0.422)			
Femaledir		0.051			
		(0.080)			
Observation	26,954	26,377	26,890	21,909	26,954
Year fixed	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.498	0.505	0.723	0.484	0.498
F	231.135	206.794	423.367	197.369	1052.899

Notes: *p < 0.1, **p < 0.05, ***p < 0.01. Robust t statistics in parentheses.

5.1. Theoretical implications

This research enriches and extends the literature on corporate digitalization by shifting the analytical focus from firm-level outcomes to supply chain governance. Prior studies predominantly conceptualize digitalization as a driver of operational efficiency and financial performance at the firm level (Abou-Foul et al., 2021; Li et al., 2022; Ribeiro-Navarrete et al., 2021; Zhu et al., 2024). A growing body of research further examines how digital technologies enhance sustainability or green performance within focal firms (Fang & Li, 2024; Karmaker et al., 2023; Zheng et al., 2024), and some studies recognize the role of digital technologies in green supply chain management from the focal firm perspective (Karmaker et al., 2023; Lerman et al., 2022). However, relatively little research directly examines how digitalization enables focal firms to actively govern and improve suppliers' environmental performance.

Our study advances this literature by conceptualizing corporate digitalization as a governance mechanism through which focal firms can coordinate, monitor, and guide suppliers' environmental practices. This perspective moves beyond viewing digitalization as a general performance-enhancing technology and instead positions it as a concrete governance infrastructure that facilitates environmental oversight and coordination across organizational boundaries. In doing so, the findings position corporate digitalization within the triple bottom line framework, highlighting its role in advancing not only economic outcomes but also environmental sustainability (Jum'a et al., 2022).

This study also contributes to the supply chain sustainability literature by offering new insights into how focal firms can proactively

Table 6
Endogeneity tests.

	Matched sample	IV-2SLS		
		First stage	Second stage	
	Escore	Digitalization	Escore	L. Digitalization
Escore				−0.000
				(−0.18)
Digitalization	0.146***		0.849**	
	(2.64)		(2.03)	
it_employee		0.154***		
		(3.11)		
Firm size	0.603***	−0.052***	0.663***	−0.063***
	(5.61)	(−2.74)	(6.49)	(−3.85)
Tobin' s *q*	0.007	−0.001	0.051	0.002
	(0.18)	(−0.12)	(1.52)	(0.33)
Capt_Intensity	−0.035	−0.033***	−0.064	−0.034***
	(−1.43)	(−3.07)	(−1.59)	(−2.96)
S_Cash	1.945**	0.564***	1.362	0.718***
	(2.28)	(3.47)	(1.60)	(3.83)
C_Cash	−0.272	1.558***	−1.633*	1.875***
	(−0.52)	(3.58)	(−1.91)	(4.25)
S_Salesgth	−0.085***	−0.000	−0.042**	0.007
	(−3.70)	(−0.06)	(−2.19)	(1.23)
C_Salesgth	−0.019**	−0.002***	−0.000	−0.001***
	(−2.10)	(−3.22)	(−0.27)	(−4.15)
C_Firmsize	−0.021	0.027	−0.068	0.078
	(−0.69)	(0.60)	(−1.25)	(1.59)
S_RD	1.308	1.932***	1.133	1.864***
	(0.58)	(3.69)	(0.65)	(2.96)
C_RD	0.828	−0.002***	−0.000	−0.001***
	(1.38)	(−13.08)	(−0.35)	(−4.82)
S_ISO9001	0.733***	−0.042	0.801***	−0.052
	(3.21)	(−1.12)	(3.76)	(−1.32)
C_ISO9001	0.281**	0.049	0.117	0.010
	(2.11)	(0.50)	(0.88)	(0.10)
S_SOE	−0.313	−0.081	−0.220	−0.135***
	(−1.64)	(−1.54)	(−1.09)	(−2.60)
C_SOE	−0.254**	−0.534***	0.196	−0.613***
	(−1.98)	(−4.08)	(0.77)	(−4.70)
Techproximity	−0.163	−0.058	−0.005	−0.048
	(−1.14)	(−0.50)	(−0.04)	(−0.39)
Same_Ind	−0.093	0.157**	−0.104	0.148**
	(−0.57)	(2.29)	(−0.61)	(2.09)
Same_Area	−0.021			0.117**
	(−0.16)			(2.17)
L.Escore	0.670***	−0.002	0.643***	0.000
	(44.47)	(−0.83)	(39.03)	(0.08)
Constant	7.111***	0.904	7.463***	1.440
	(3.15)	(0.79)	(2.93)	(1.26)
Year fixed	Yes	Yes	Yes	Yes
Observations	13,146	21,009	21,009	20,341
R-squared	0.525	0.182	0.494	0.190

Notes: *p < 0.1, **p < 0.05, ***p < 0.01. Robust *t* statistics in parentheses.

manage suppliers' environmental performance. Existing research emphasizes that improving a firm's own sustainability performance is insufficient, as environmental misconduct by supply chain partners can generate substantial reputational and regulatory risks for focal firms (Lo et al., 2018; Pizzetti et al., 2021). Building on this stream of research, our study moves beyond documenting spillover effects and instead provides a governance-oriented explanation of how focal firms can actively shape suppliers' environmental behavior. Rather than treating environmental improvements as passive spillover outcomes, we demonstrate that focal firm digitalization constitutes a scalable governance mechanism through which suppliers' environmental practices can be systematically monitored and enhanced. In this way, our study shifts the focus of supply chain sustainability research from explaining why environmental risks propagate to clarifying how focal firms can prevent and mitigate such risks through digitally enabled governance.

Finally, this research contributes to both the digitalization and supply chain sustainability literatures by incorporating institutional theory as a contextual lens to explain when and how digital governance mechanisms are most effective. Prior studies on corporate digitalization commonly draw on the resource-based view (Zhu et al., 2024), information processing theory (Zhang et al., 2022), or the dynamic capabilities perspective (Meier et al., 2023). While these frameworks offer valuable insights into internal capability development, they provide limited explanation of how external institutional environments shape the governance implications of digitalization across supply chains. Drawing on institutional theory, we argue that institutional pressures influence focal firms' incentives to deploy, intensify, and enforce digital governance mechanisms. Specifically, we develop a framework in which coercive, mimetic, and normative pressures condition the effectiveness of focal firm digitalization in managing suppliers' environmental practices. This perspective underscores the importance of incorporating institutional contingencies when evaluating the sustainability implications of digital transformation and advances institutional theory by clarifying how institutional mechanisms interact with digital governance to shape interorganizational environmental outcomes.

5.2. Practical implications

This research offers practical implications for supply chain-level environmental management. For focal firms operating in an era of heightened environmental scrutiny, strategically deploying digitalization to improve supplier environmental performance has become increasingly critical. The concept of a green supply chain emphasizes the integration of sustainability considerations throughout the entire supply chain (Lerman et al., 2022). Our findings suggest that focal firms should proactively leverage digitalization to govern suppliers' sustainability practices, including aligning suppliers with environmental objectives, streamlining monitoring and compliance processes, and automating or optimizing resource allocation decisions. More importantly, digitalization can foster deeper green collaboration between focal firms and suppliers. With the rapid advancement of digital technologies such as artificial intelligence and big data analytics, focal firms should continuously explore innovative approaches (e.g., AI-enabled joint energy-efficiency initiatives) to collaborate with suppliers and further strengthen supply chain sustainability.

For suppliers, understanding the complexity of the institutional environment is equally essential, as their actions are accountable not only to direct regulatory institutions but also to broader institutional forces indirectly connected to them. Such awareness enables suppliers to more effectively interpret the pressures and incentives emerging from multiple sources, thereby reducing the risk of overlooking salient expectations embedded in the broader institutional context. In particular, suppliers should carefully consider the environmental requirements imposed by focal firms, including the focal firms' regulatory environments, prevailing industry practices, and expectations associated with international market exposure. By developing a more comprehensive understanding of these institutional dimensions, suppliers can better mobilize the resources and capabilities necessary to address both short- and long-term environmental challenges.

5.3. Limitations and future research directions

This study has several limitations that warrant further investigation. First, consistent with prior research, our measure of digitalization intensity captures the frequency of more than 100 digitalization-related keywords without distinguishing among specific categories. Future research could extend our work by classifying these keywords to capture different dimensions or functional types of digitalization, such as operational versus strategic adoption, internal versus external process digitalization, and green-oriented versus efficiency-oriented applications. Such refinements would facilitate a more nuanced understanding of heterogeneous digitalization strategies and their distinct performance

implications.

Second, our analysis is conducted within the Chinese context, which may constrain the generalizability of our findings. The focal firm-supplier relationships examined in this study are embedded in a specific institutional environment, and the observed effects may differ in contexts characterized by greater geographic dispersion or alternative institutional configurations. Future research could investigate firms operating across multiple countries to better understand how institutional heterogeneity shapes the proposed relationships.

Finally, this study focuses on the role of focal firm digitalization as a governance mechanism in shaping the environmental performance of tier-1 suppliers. However, digitally enabled governance effects may extend beyond direct supplier relationships. Future research could explore multi-tier supply chain structures, as the increasing complexity of supply chain networks may influence how digital governance mechanisms operate and ultimately affect overall supply chain sustainability.

CRediT authorship contribution statement

Miaomiao Zhang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peter K.C. Lee:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization. **Chen Liang:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Minghao Zhu:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Conceptualization. **Andy C. L. Yeung:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **T.C.E. Cheng:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Abou-Foul, M., Ruiz-Alba, J. L., & Soares, A. (2021). The impact of digitalization and servitization on the financial performance of a firm: An empirical analysis. *Production Planning & Control, 32*(12), 975–989.

Agca, S., Babich, V., Birge, J. R., & Wu, J. (2022). Credit shock propagation along supply chains: Evidence from the CDS market. *Management Science, 68*(9), 6506–6538.

Angell, L. C., & Klassen, R. D. (1999). Integrating environmental issues into the mainstream: An agenda for research in operations management. *Journal of Operations Management, 17*(5), 575–598.

Benitez, G. B., Ferreira-Lima, M., Ayala, N. F., & Frank, A. G. (2022). Industry 4.0 technology provision: The moderating role of supply chain partners to support technology providers. *Supply Chain Management: An International Journal, 27*(1), 89–112.

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics, 29*(2), 238–249.

Campbell, J. L. (2007). Why would corporations behave in socially responsible ways? An institutional theory of corporate social responsibility. *Academy of Management Review, 32*(3), 946–967.

Castro-Lopez, A., Iglesias, V., & Santos-Vijande, M. L. (2023). Organizational capabilities and institutional pressures in the adoption of circular economy. *Journal of Business Research, 161*, Article 113823.

Chang, H., Liu, S., & Mashruwala, R. (2022). Customer bargaining power, strategic fit, and supplier performance. *Production and Operations Management, 31*(4), 1492–1509.

Chen, S., Lai, K. H., Guo, X., & Zhang, X. (2025). The influence of digital health technology on the allocation of regional medical resources in China. *Health Policy and Technology, 14*(3), Article 101013.

Chen, X., Chen, L., Jiang, M., & Yan, J. (2021). Does R&D intensity promote the adoption of circular supply chain management? Evidence from China. *Industrial Marketing Management, 99*, 153–166.

Choi, T. M., Kumar, S., Yue, X., & Chan, H. L. (2022). Disruptive technologies and operations management in the Industry 4.0 era and beyond. *Production and Operations Management, 31*(1), 9–31.

Chu, Z., Xu, J., Lai, F., & Collins, B. J. (2018). Institutional theory and environmental pressures: The moderating effect of market uncertainty on innovation and firm performance. *IEEE Transactions on Engineering Management, 65*(3), 392–403.

Corbett, C. J., & Klassen, R. D. (2006). Extending the horizons: Environmental excellence as key to improving operations. *Manufacturing & Service Operations Management, 8*(1), 5–22.

Dalenogare, L. S., Benitez, G. B., Ayala, N. F., & Frank, A. G. (2018). The expected contribution of Industry 4.0 technologies for industrial performance. *International Journal of Production Economics, 204*, 383–394.

Danese, P., Lion, A., & Vinelli, A. (2019). Drivers and enablers of supplier sustainability practices: A survey-based analysis. *International Journal of Production Research, 57*(7), 2034–2056.

Deng, X., Li, W., & Ren, X. (2023). More sustainable, more productive: Evidence from ESG ratings and total factor productivity among listed Chinese firms. *Finance Research Letters, 51*, Article 103439.

DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review, 48*(2), 147–160.

Escobar, L. F., & Vredenburg, H. (2011). Multinational oil companies and the adoption of sustainable development: A resource-based and institutional theory interpretation of adoption heterogeneity. *Journal of Business Ethics, 98*(1), 39–65.

Fang, L., & Li, Z. (2024). Corporate digitalization and green innovation: Evidence from textual analysis of firm annual reports and corporate green patent data in China. *Business Strategy and the Environment, 33*(5), 3936–3964.

Frank, A. G., Dalenogare, L. S., & Ayala, N. F. (2019). Industry 4.0 technologies: Implementation patterns in manufacturing companies. *International Journal of Production Economics, 210*, 15–26.

Feng, Y., & Zhu, Q. (2024). How do customers' environmental efforts diffuse to suppliers: The role of customers' characteristics and suppliers' digital technology capability. *International Journal of Operations & Production Management, 44*(9), 1676–1706.

Galbreth, M. R., & Ghosh, B. (2013). Competition and sustainability: The impact of consumer awareness. *Decision Sciences, 44*(1), 127–159.

Geng, R., Lam, H. K. S., & Stevenson, M. (2022). Addressing modern slavery in supply chains: An awareness-motivation-capability perspective. *International Journal of Operations & Production Management, 42*(3), 331–356.

Genovese, A., Lenny Koh, S., Kumar, N., & Tripathi, P. K. (2014). Exploring the challenges in implementing supplier environmental performance measurement models: A case study. *Production Planning & Control, 25*(13–14), 1198–1211.

Ghadge, A., Jena, S. K., Kamble, S., Misra, D., & Tiwari, M. K. (2021). Impact of financial risk on supply chains: A manufacturer-supplier relational perspective. *International Journal of Production Research, 59*(23), 7090–7105.

Gong, M., Gao, Y., Koh, L., Sutcliffe, C., & Cullen, J. (2019). The role of customer awareness in promoting firm sustainability and sustainable supply chain management. *International Journal of Production Economics, 217*, 88–96.

Govindan, K., & Hasanagic, M. (2018). A systematic review on drivers, barriers, and practices towards circular economy: A supply chain perspective. *International Journal of Production Research, 56*(1–2), 278–311.

Hartmann, J., & Uhlenbruck, K. (2015). National institutional antecedents to corporate environmental performance. *Journal of World Business, 50*(4), 729–741.

Jia, F., Xu, Y., Chen, L., & Fernandes, K. (2025). Does supply chain concentration improve sustainability performance: The role of operational slack and information transparency. *International Journal of Operations & Production Management, 45*(1), 269–300.

Jum'a, L., Zimon, D., Ikram, M., & Madzik, P. (2022). Towards a sustainability paradigm; The nexus between lean green practices, sustainability-oriented innovation and Triple Bottom Line. *International Journal of Production Economics, 245*, Article 108393.

Kamble, S. S., & Gunasekaran, A. (2023). Analysing the role of Industry 4.0 technologies and circular economy practices in improving sustainable performance in Indian manufacturing organisations. *Production Planning & Control, 34*(10), 887–901.

Karmaker, C. L., Al Aziz, R., Ahmed, T., Misbauddin, S., & Moktadir, M. A. (2023). Impact of industry 4.0 technologies on sustainable supply chain performance: The mediating role of green supply chain management practices and circular economy. *Journal of Cleaner Production, 419*, Article 138249.

Kauppi, K. (2013). Extending the use of institutional theory in operations and supply chain management research: Review and research suggestions. *International Journal of Operations & Production Management, 33*(10), 1318–1345.

Kleibergen, F., & Paap, R. (2006). Generalized reduced rank tests using the singular value decomposition. *Journal of Econometrics, 133*(1), 97–126.

Laari, S., Töyli, J., Solakivi, T., & Ojala, L. (2016). Firm performance and customer-driven green supply chain management. *Journal of Cleaner Production, 112*, 1960–1970.
Lerman, L. V., Benitez, G. B., Müller, J. M., de Sousa, P. R., & Frank, A. G. (2022). Smart green supply chain management: A configurational approach to enhance green performance through digital transformation. *Supply Chain Management: An International Journal, 27*(7), 147–176.
Li, L., Ye, F., Zhan, Y., Kumar, A., Schiavone, F., & Li, Y. (2022). Unraveling the performance puzzle of digitalization: Evidence from manufacturing firms. *Journal of Business Research, 149*, 54–64.
Li, M., Cao, G., Cui, L., Liu, X., & Dai, J. (2023). Examining how government subsidies influence firms' circular supply chain management: The role of eco-innovation and top management team. *International Journal of Production Economics, 261*, Article 108893.
Li, Y., Dai, J., & Cui, L. (2020). The impact of digital technologies on economic and environmental performance in the context of industry 4.0: A moderated mediation model. *International Journal of Production Economics, 229*, Article 107777.
Liang, C., Lee, P. K. C., Zhu, M., Yeung, A. C. L., Cheng, T. C. E., & Zhou, H. (2024). The bright side of being uncertain: The impact of economic policy uncertainty on corporate innovation. *International Journal of Operations & Production Management, 44*(11), 1886–1913.
Lin, X., Zhu, H., & Meng, Y. (2023). ESG greenwashing and equity mispricing: Evidence from China. *Finance Research Letters, 58*, Article 104606.
Liu, H., Ke, W., Wei, K. K., Gu, J., & Chen, H. (2010). The role of institutional pressures and organizational culture in the firm's intention to adopt internet-enabled supply chain management systems. *Journal of Operations Management, 28*(5), 372–384.
Liu, W., Chen, Z., Choi, T.-M., Lee, P.-T.-W., Chan, H. K., & Gao, Y. (2024). Carbon neutral announcements and Chinese stock market reaction: A supply chain network-based perspective. *International Journal of Operations & Production Management, 44*(4), 729–767.
Lo, C. K. Y., Tang, C. S., Zhou, Y., Yeung, A. C. L., & Fan, D. (2018). Environmental incidents and the market value of firms: An empirical investigation in the Chinese context. *Manufacturing & Service Operations Management, 20*(3), 422–439.
Mateska, I., Busse, C., Kach, A. P., & Wagner, S. M. (2023). Sustainability-related transgressions in global supply chains: When do legitimacy spillovers hurt buying firms the most? *Journal of Supply Chain Management, 59*(4), 42–78.
Meier, O., Gruchmann, T., & Ivanov, D. (2023). Circular supply chain management with blockchain technology: A dynamic capabilities view. *Transportation Research Part E: Logistics and Transportation Review, 176*, Article 103177.
Meindl, B., Ayala, N. F., Mendonça, J., & Frank, A. G. (2021). The four smarts of Industry 4.0: Evolution of ten years of research and future perspectives. *Technological Forecasting and Social Change, 168*, Article 120784.
Mena, C., & Schoenherr, T. (2023). The green contagion effect: An investigation into the propagation of environmental practices across multiple supply chains tiers. *International Journal of Production Research, 61*(14), 4808–4825.
Nasiri, M., Ukko, J., Saunila, M., & Rantala, T. (2020). Managing the digital supply chain: The role of smart technologies. *Technovation, 96*, Article 102121.
Park, J., Han, K., & Lee, B. (2023). Green cloud? An empirical analysis of cloud computing and energy efficiency. *Management Science, 69*(3), 1639–1664.
Pizzetti, M., Gatti, L., & Seele, P. (2021). Firms talk, suppliers walk: Analyzing the locus of greenwashing in the blame game and introducing 'vicarious greenwashing'. *Journal of Business Ethics, 170*(1), 21–38.
Porteous, A. H., Rammohan, S. V., & Lee, H. L. (2015). Carrots or sticks? Improving social and environmental compliance at suppliers through incentives and penalties. *Production and Operations Management, 24*(9), 1402–1413.
Rao, P. (2002). Greening the supply chain: A new initiative in South East Asia. *International Journal of Operations & Production Management, 22*(6), 632–655.
Ribeiro-Navarrete, S., Botella-Carrubi, D., Palacios-Marqués, D., & Orero-Blat, M. (2021). The effect of digitalization on business performance: An applied study of KIBS. *Journal of Business Research, 126*, 319–326.
Sarkis, J., Gonzalez-Torre, P., & Adenso-Diaz, B. (2010). Stakeholder pressure and the adoption of environmental practices: The mediating effect of training. *Journal of Operations Management, 28*(2), 163–176.
Scott, W. R. (2008). *Institutions and Organizations: Ideas and Interests*.
Shafiq, A., Johnson, P. F., Klassen, R. D., & Awaysheh, A. (2017). Exploring the implications of supply risk on sustainability performance. *International Journal of Operations & Production Management, 37*(10), 1386–1407.
Song, S., Dong, Y., Kull, T., Carter, C., & Xu, K. (2023). Supply chain leakage of greenhouse gas emissions and supplier innovation. *Production and Operations Management, 32*(3), 882–903.
Song, S., Lian, J., Skowronski, K., & Yan, T. (2024). Customer base environmental disclosure and supplier greenhouse gas emissions: A signaling theory perspective. *Journal of Operations Management, 70*(3), 355–380.
Stock, J. H., & Yogo, M. (2002). Testing for weak instruments in linear IV regression. *NBER Technical Working Paper 0284*.
Sun, P., Qu, Z., & Liao, Z. (2018). How and when do subnational institutions matter for R&D investment? evidence from the Chinese pharmaceutical sector. *IEEE Transactions on Engineering Management, 65*(3), 379–391.
Talluri, S., & Sarkis, J. (2002). A model for performance monitoring of suppliers. *International Journal of Production Research, 40*(16), 4257–4269.
Turkulainen, V., Kauppi, K., & Nermes, E. (2017). Institutional explanations: Missing link in operations management? Insights on supplier integration. *International Journal of Operations & Production Management, 37*(8), 1117–1140.
Villena, V. H., & Dhanorkar, S. (2020). How institutional pressures and managerial incentives elicit carbon transparency in global supply chains. *Journal of Operations Management, 66*(6), 697–734.
Walls, J. L., Berrone, P., & Phan, P. H. (2012). Corporate governance and environmental performance: Is there really a link? *Strategic Management Journal, 33*(8), 885–913.
Wang, J., Qi, B., Li, Y., Hossain, M. I., & Tian, H. (2024). Does institutional commitment affect ESG performance of firms? Evidence from the United Nations principles for responsible investment. *Energy Economics, 130*, Article 107302.
Wen, X., Cheah, J.-H., Lim, X.-J., & Ramachandran, S. (2023). Why does "green" matter in supply chain management? Exploring institutional pressures, green practices, green innovation, and economic performance in the Chinese chemical sector. *Journal of Cleaner Production, 427*, Article 139182.
Wilhelm, M. M., Blome, C., Bhakoo, V., & Paulraj, A. (2016). Sustainability in multi-tier supply chains: Understanding the double agency role of the first-tier supplier. *Journal of Operations Management, 41*(1), 42–60.
Wu, X. (2024). *Apple Must Clean up Its Polluting Supply Chain – Starting With Foxconn*. Retrieved from https://thediplomat.com/2024/01/apple-must-clean-up-its-polluting-supply-chain-starting-with-foxconn/.
Xu, Y., Chin, W., Liu, Y., & He, K. (2022). Do institutional pressures promote green innovation? The effects of cross-functional coopetition in green supply chain management. *International Journal of Physical Distribution & Logistics Management, 53*(7–8), 743–761.
Yang, Y., Lau, A. K., Lee, P. K., Yeung, A. C., & Cheng, T. E. (2019). Efficacy of China's strategic environmental management in its institutional environment. *International Journal of Operations & Production Management, 39*(1), 138–163.
Yang, Z., Hu, W., Shao, J., Shou, Y., & He, Q. (2023). How does digitalization alter the paradox of supply base concentration? The effects of digitalization intensity and breadth. *International Journal of Operations & Production Management, 43*(11), 1690–1722.
Ye, F., Huang, G., Zhan, Y., & Li, Y. (2021). Factors mediating and moderating the relationships between green practice and environmental performance: Buyer–supplier relation and institutional context. *IEEE Transactions on Engineering Management, 70*(1), 142–155.
Ye, F., Ouyang, Y., & Li, Y. (2023). Digital investment and environmental performance: The mediating roles of production efficiency and green innovation. *International Journal of Production Economics, 259*, Article 108822.
Zhang, C., Venkatesh, M., & Ohana, M. (2024). The impact of normative institutions on socially sustainable supply chain management: The role of individual cultural values. *International Journal of Operations & Production Management, 44*(4), 790–812.
Zhang, X., Pu, J., Lu, Y., & Guo, F. (2025). Team makes you better: Evidence from online medical consultation platforms. *Information Systems Research, 36*(3), 1738–1759.
Zhang, Y., Huo, B., Haney, M. H., & Kang, M. (2022). The effect of buyer digital capability advantage on supplier unethical behavior: A moderated mediation model of relationship transparency and relational capital. *International Journal of Production Economics, 253*, Article 108603.
Zheng, D., Song, H., Zhao, C., Liu, Y., & Zhao, W. (2024). Is it possible for semiconductor companies to reduce carbon emissions through digital transformation? Evidence from China. *International Journal of Production Economics, 272*, Article 109246.
Zhou, X., Ying, S. X., You, J., & Wu, H. (2024). Like parent, like child: MNCs' CSR and their foreign subsidiaries' environmental footprint. *Journal of Business Research, 172*, Article 114413.
Zhu, M., Liang, C., Yeung, A. C. L., & Zhou, H. (2024). The impact of intelligent manufacturing on labor productivity: An empirical analysis of Chinese listed manufacturing companies. *International Journal of Production Economics, 267*, Article 109070.
Zhu, Q., Cordeiro, J., & Sarkis, J. (2012). International and domestic pressures and responses of Chinese firms to greening. *Ecological Economics, 83*, 144–153.
Zhu, Q., & Sarkis, J. (2007). The moderating effects of institutional pressures on emergent green supply chain practices and performance. *International Journal of Production Research, 45*(18–19), 4333–4355.
Zimmer, K., Fröhling, M., & Schultmann, F. (2016). Sustainable supplier management–A review of models supporting sustainable supplier selection, monitoring and development. *International Journal of Production Research, 54*(5), 1412–1442.

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